Short Curriculum Vitæ and list of publications

Dr. Fantoni Riccardo
August 30, 1970
Italian
phone: +39-040-43372
cell.: +393384570334
e-mail: rfantoni@ts.infn.it
homepage: http://www-dft.ts.infn.it/∼rfantoni/

Education

1984/1989 "Maturità" 60/60 at Liceo Scientifico Sperimentale Filippo Buonarroti in Pisa.

1989/1994 November 18th "Laurea" Degree in Physics 110/110 cum laude at the University of Pisa, Italy.

1995/1997 May 18th Master Degree in Physics at the University of Illinois at Urbana-Champaign, U.S.A..

2001/2004 April 6th Philosophical Degree in Physics at the University of Trieste, Italy.

Positions

1996 summer: Research Assistant in the Physics department at the University of Illinois at Urbana-Champaign. Coordinators David Ceperley and Richard Martin.

1997 summer: Research Assistant in the Physics department at the University of Illinois at Urbana-Champaign. Coordinators David Ceperley and Richard Martin.

1997 fall: Research Assistant in the Physics department at the University of Illinois at Urbana-Champaign. Coordinators David Ceperley and Richard Martin.

1999 August 25th: Traditional Term Cancellation from the Ph.D. program at the University of Illinois at Urbana-Champaign. Earned Hrs 106, GPA Hrs 106, Points 388, GPA 3.66 (Total number of hours required for a Ph.D.: 96).

2000 Won a Ph.D. position (BAT-O IIa/2) at the Max-Planck-Institute für Plasmaphysik, Teilinstitut Greifswald, Bereich Stellaratortheorie at the 5th of January 2000 which I declined.

2000 Research Assistant (1 year) at the Department of Mathematics and Statistics, University of Limerick, Ireland in the group of S. B. G. O'Brien.

2003 January, Visiting Scientist at the University of Paris-Sud at Orsay. Scientific collaboration with the research group of Prof. Bernard Jancovici.

2004 April 15th, "Assegno di ricerca" (Postdoctoral Fellowship 24 months) at the Department of Chemical Physics, University "Ca' Foscari" of Venice. Title of the project: "Analysis of proteins in solutions using statistical thermodynamic techniques". Project PRIN-COFIN2003025755-044. National coordinator Prof. Amos Maritan. Local coordinator Prof. Achille Giacometti.

2006 March 17th, "Assegno di ricerca" (Postdoctoral Fellowship 24 months) at the Department of Chemical Physics, University "Ca' Foscari" of Venice. Title of the project: "Colloidal mixtures, globular proteins and liquid crystal-like phases of biopolymers". Project PRIN-COFIN2005027330. National coordinator Prof. Amos Maritan. Local coordinator Prof. Achille Giacometti.

2009 October 15th, Postdoctoral Fellowship (36 months) at the National Institute of Theoretical Physics (NITheP), Stellenbosch Institute for Advanced Study, Matieland 7602, South Africa.

2010 October 10th, Visiting scientist (three months) at the University of Extremadura in Badajoz, Spain. Scientific collaboration with the research group of Prof. Andrés Santos.

2012 January 1st, Guest (three years) at S.I.S.S.A., via Bonomea 265, 34136 Trieste, Italy.

2012 January 1st, Honorary Fellow at the Department of Molecular Science and Nanosystems, University "Ca' Foscari" of Venice, Calle Larga S. Marta DD2137, I-30123 Venezia, Italy.

2015 July 17th: Habilitation to teach Mathematics and Physics (classes A049, A048, A047, A038) in Italian high-school obtained at the University of Trieste with mark 87/100. One year course with exams. "Abilitazione all' insegnamento nella scuola secondaria per la classe A049 Matematica e Fisica con punteggio 87/100".

2016 August 24th: Passed a "Concorso per l' assunzione in ruolo nell' insegnamento di Matematica (classe A026) nella scuola secondaria di secondo grado con punteggio 72.2/100".

2017 September 1st: "Contratto a tempo indeterminato per l' insegnamento di Matematica nella scuola secondaria di secondo grado (confirmed on 1st September 2019)".

2019 May 10th: Habilitation for Associate Professor in Theoretical Physics of Matter (sector 02/B2) in the Italian University (valid until 10 May 2030).

Short visits and scientific collaborations

1995-2000 Collaboration in the Physics Department at the University of Illinois at Urbana-Champaign, USA with the research group of Prof. David Ceperley and Prof. Richard Martin.

2000-2001 Collaboration with the Research group of Prof. S. B. G. O'Brien at the Department of Mathematics and Statistics, University of Limerick, Ireland.

2001-2004 Collaboration with the research group of Prof. Giorgio Pastore at the Theoretical Physics Department of the University of Trieste, Italy.

2003 January: Visiting Scientist (1 month) at the University of Paris-Sud at Orsay. Scientific collaboration with the research group of Prof. Bernard Jancovici at the Laboratory of Theoretical Physics of the University of Paris-Sud at Orsay, France.

2004-2006 Title of the project: "Analysis of proteins in solutions using statistical thermodynamic techniques". Project PRIN-COFIN2003025755-044. University "Ca' Foscari" of Venice, Italy. National coordinator Prof. Amos Maritan. Local coordinator Prof. Achille Giacometti.

2006-2008 Title of the project: "Colloidal mixtures, globular proteins and liquid crystal-like phases of biopolymers". Project PRIN-COFIN2005027330. University "Ca' Foscari" of Venice, Italy. National coordinator Prof. Amos Maritan. Local coordinator Prof. Achille Giacometti.

2009 Research group of Prof. Michael Kastner at the National Institute of Theoretical Physics of the University of Stellenbosch, South Africa.

2009-2011 Research group of Prof. K. K. Müller-Nedebock in the Physics Department at the University of Stellenbosch, South Africa.

2010 October 10th: Visiting scientist (three months) at the University of Extremadura in Badajoz, Spain. Scientific collaboration with the research group of Prof. Andrés Santos in the Physics Department at the University of Extremadura in Badajoz, Spain.

2011 Research group of Prof. Bert Klumperman in the Polymer Physics Department at the University of Stellenbosch, South Africa.

2013 October 20th: Visiting Scientist (one month) at the University of Extremadura in Badajoz, Spain. Scientific collaboration with the research group of Prof. Andrés Santos in the Physics Department at the University of Extremadura in Badajoz, Spain.

2014 Research group of Prof. Saverio Moroni in Scuola Internazionale Superiore di Studi Avanzati (SISSA), Italy.

2014 June 15th: Visiting Scientist (one week) at the "Laboratoire de Physique ENS de Lyon", France. Scientific collaboration with the research group of Prof. Angel Alastuey.

2017 January 8th: Visiting Scientist (two months) at the University of Extremadura in Badajoz, Spain. Scientific collaboration with the research group of Prof. Andrés Santos in the Physics Department at the University of Extremadura in Badajoz, Spain.

2018 March 26th: Visiting Scientist (one week) at the "Laboratoire de Physique ENS de Lyon", France. Scientific collaboration with the research group of Prof. Angel Alastuey.

Research Interests

Aim of the research is to develop analytical and computational methods for condensed and

soft matter starting from the fundamental many-body equations. Apart from the few analytically exactly solvable models our principal instruments are Integral Equation Theory, Density Functional Theory, Thermodynamic Perturbation Theory, Association Theory, and Monte Carlo simulations which can find exact properties of many-body systems. We are combining these approaches to create new methods and to test the accuracy of calculations on materials. Current studied materials include colloidal suspensions, ionic liquids, polymer mixtures, the electron fluid, the polaron, and boson fluids (like ^{4}He, ^{4}He-H_2 mixtures, ...). We investigate the structure and thermodynamic properties of the materials including their phase transitions like the gas-liquid-solid first order ones, the percolation threshold, the clustering, the localization, the demixing, the polydispersity, and surface properties.
Lately I started working on euclidean relativistic covariant and ultralocal quantum scalar field theories through Path integral Monte Carlo of lattice field theory subject to different kinds of quantization procedures.

Teaching

1995/2000 Teaching Assistant in the Physics department at the University of Illinois at Urbana-Champaign (Electricity and Magnetism I and II, Biomolecular Physics, Waves in Physics, Thermal Physics and Fluids, Classical Mechanics, Electricity, Magnetism, and Optics, Waves and Quantum mechanics/Thermal Physics and Fluids).

2004/2006 Teaching at University Ca' Foscari of Venice (Mathematical Methods for Science and Technlogy of Materials, Analysis II).

2008/2009 Teaching at University of Trieste (General Physics I fo Industrial Engineering).

2012/2013 Teaching Assistant at University of Trieste (Laboratory of Optics).

2013/2014 Teaching Assistant at University of Trieste (Laboratory of Calculus).

2016/2017 Teaching Assistant at University of Trieste (Laboratory of Calculus).

Publications

I have **78 publications** with an **h-index of 16**, a total sum of the **times cited of 745**, **27** publications where I am the **only author**, and **40** publications where I am the **first author**. These data are extracted from the ISI database on December 2023. Follows the list of publications:

(1) Fantoni R. and Tosi M.P., Nuovo Cimento **17D**, 155 (1995)
dx.doi.org/10.1007/BF02451594
WOS:A1995QW87300005
2-s2.0-51649140258

(2) Fantoni R. and Tosi M.P., Nuovo Cimento **17D**, 1165 (1995)
dx.doi.org/10.1007/BF02454131

WOS:A1995TM36400009
2-s2.0-51649138270

(3) Fantoni R. and Tosi M.P., Physica B **217**, 35 (1996)
dx.doi.org/10.1016/0921-4526(95)00451-3
WOS:A1996TR03900005
2-s2.0-0030561994

(4) Fantoni R., Jancovici B., and Téllez G., J. Stat. Phys. **112**, 27 (2003)
dx.doi.org/10.1023/A:1023671419021
WOS:000182711300002
2-s2.0-0037726817

(5) Fantoni R. and Pastore G., J. Chem. Phys. **119**, 3810 (2003)
dx.doi.org/10.1063/1.1590642
WOS:000184474100028
2-s2.0-0041376961

(6) Fantoni R. and Pastore G., Phys. Rev. E **68**, 046104 (2003)
dx.doi.org/10.1103/PhysRevE.68.046104
WOS:000186571200013
2-s2.0-0347566148

(7) Fantoni R. and Pastore G., Physica A **332**, 349 (2004)
dx.doi.org/10.1016/j.physa.2003.10.012
WOS:000188086200022
2-s2.0-0346215805

(8) Fantoni R. and Pastore G., J. Chem. Phys. **120**, 10681 (2004)
dx.doi.org/10.1063/1.1739392
WOS:000221538200034
2-s2.0-2942662065

(9) Fantoni R., Gazzillo D., Giacometti A., J. Chem. Phys. **122**, 034901 (2005)
dx.doi.org/10.1063/1.1831275
WOS:000226748800038
2-s2.0-22944457598

(10) Fantoni R., Gazzillo D., Giacometti A., Phys. Rev. E **72**, 011503 (2005)
dx.doi.org/10.1103/PhysRevE.72.011503
WOS:000230886900043
2-s2.0-27244434158

(11) Fantoni R., Gazzillo D., Giacometti A., and Sollich P. J. Chem. Phys. **125**, 164504 (2006)
dx.doi.org/10.1063/1.2358136
WOS:000241722000048
2-s2.0-33750439815

(12) Gazzillo D., Giacometti A., **Fantoni R.**, and Sollich P., Phys. Rev. E. **74**, 051407 (2006)
dx.doi.org/10.1103/PhysRevE.74.051407
WOS:000242408700047
2-s2.0-33751364294

(13) Gazzillo D., **Fantoni R.**, and Giacometti A., Mol. Phys. **104**, 3451 (2006)
dx.doi.org/10.1080/00268970601050892
WOS:000243977700009
2-s2.0-34547821930

(14) **Fantoni R.**, Gazzillo D., Giacometti A., Miller M. A., and Pastore G., J. Chem. Phys. **127**, 234507 (2007)
dx.doi.org/10.1063/1.2805066
WOS:000251908500033
2-s2.0-37549062945

(15) Gazzillo D., **Fantoni R.**, and Giacometti A., Phys. Rev. E **78**, 021201 (2008)
dx.doi.org/10.1103/PhysRevE.78.021201
WOS:000259263600051
2-s2.0-49549117568

(16) Santos A., **Fantoni R.**, and Giacometti A., Phys. Rev. E **77**, 051206 (2008)
dx.doi.org/10.1103/PhysRevE.77.051206
WOS:000256885400051
2-s2.0-44149108199

(17) **Fantoni R.** and Téllez G., J. Stat. Phys. **133**, 449 (2008)
dx.doi.org/10.1007/s10955-008-9616-x
WOS:000260376300004
2-s2.0-54949158394

(18) **Fantoni R.**, Giacometti A., Malijevský A., and Santos A., J. Chem. Phys. **131**, 124106 (2009)
dx.doi.org/10.1063/1.3236515
WOS:000270380300010
2-s2.0-70349617808

(19) Gazzilo D., **Fantoni R.**, and Giacometti A., Phys Rev. E **80**, 061207 (2009)
dx.doi.org/10.1103/PhysRevE.80.061207
WOS:000273227500061
2-s2.0-75349098936

(20) Santos A., **Fantoni R.**, and Giacometti A., J. Chem. Phys. **131**, 181105 (2009)
dx.doi.org/10.1063/1.3265991
WOS:000272454500005
2-s2.0-72949085930

(21) **Fantoni R.**, Giacometti A., Malijevský A., and Santos A., J. Chem. Phys. **133**, 024101 (2010)
dx.doi.org/10.1063/1.3455330

WOS:000279917700003
2-s2.0-77955754953

(22) **Fantoni R.**, J. Stat. Mech. P07030 (2010)
dx.doi.org/10.1088/1742-5468/2010/07/P07030
WOS:000281744400004
2-s2.0-77957073614

(23) **Fantoni R.**, A. Giacometti, F. Sciortino, and G. Pastore, Soft Matter **7**, 2419 (2011)
dx.doi.org/10.1039/C0SM00995D
WOS:000288162500026
2-s2.0-79952395236

(24) **Fantoni R.**, A. Malijevský, A. Santos, and A. Giacometti, Europhys. Lett. **93**, 26002 (2011)
dx.doi.org/10.1209/0295-5075/93/26002
WOS:000287356100015
2-s2.0-79952604983

(25) **Fantoni R.**, A. Malijevský, A. Santos, and A. Giacometti, Mol. Phys. **109**, 2723 (2011)
dx.doi.org/10.1080/00268976.2011.597357
WOS:000299109300004
2-s2.0-84855979779

(26) **Fantoni R.** and K. K. Müller-Nedebock, Phys. Rev. E **84**, 011808 (2011)
dx.doi.org/10.1103/PhysRevE.84.011808
WOS:000293457100008
2-s2.0-79961114703

(27) **Fantoni R.** and Santos A., Phys. Rev. E **84**, 041201 (2011)
dx.doi.org/10.1103/PhysRevE.84.041201
WOS:000296518800003
2-s2.0-80054919881

(28) **Fantoni R.**, Eur. Phys. J. B **85**, 108 (2012)
dx.doi.org/10.1140/epjb/e2012-20820-1
WOS:000302642200018
2-s2.0-85007085374

(29) **Fantoni R.**, J. Stat. Mech. P04015 (2012)
dx.doi.org/10.1088/1742-5468/2012/04/P04015
WOS:000303545700017
2-s2.0-84860526515

(30) **Fantoni R.**, Salari J. W. O., Klumperman B., Phys. Rev. E **85**, 061404 (2012)
dx.doi.org/10.1103/PhysRevE.85.061404
WOS:000304858000003
2-s2.0-84862224594

(31) **Fantoni R.**, Phys. Rev. B **86**, 144304 (2012)
dx.doi.org/10.1103/PhysRevB.86.144304
WOS:000309578100003
2-s2.0-84867308368

(32) **Fantoni R.**, J. Stat. Mech. P10024 (2012)
dx.doi.org/10.1088/1742-5468/2012/10/P10024
WOS:000310585500027
2-s2.0-84868286397

(33) **Fantoni R.**, Physica B **412**, 112 (2013)
dx.doi.org/10.1016/j.physb.2012.12.032
WOS:000314764900023
2-s2.0-84873386313

(34) **Fantoni R.**, Solid State Communications **159**, 106 (2013)
dx.doi.org/10.1016/j.ssc.2013.01.028
WOS:000317800900025
2-s2.0-84875448172

(35) **Fantoni R.** and Pastore G., Europhys. Lett. **101**, 46003 (2013)
dx.doi.org/10.1209/0295-5075/101/46003
WOS:000315999100021
2-s2.0-84874834405

(36) Maestre M. A. G., **Fantoni R.**, Giacometti A. and Santos A., J. Chem. Phys. **138**, 094904 (2013)
dx.doi.org/10.1063/1.4793626
WOS:000315874200054
2-s2.0-84874926774

(37) **Fantoni R.** and Santos A., Phys. Rev. E **87**, 042102 (2013)
dx.doi.org/10.1103/PhysRevE.87.042102
WOS:000317104900005
2-s2.0-84876703163

(38) **Fantoni R.** and Pastore G., Phys. Rev. E **87**, 052303 (2013)
dx.doi.org/10.1103/PhysRevE.87.052303
WOS:000319061000006
2-s2.0-84878389283

(39) **Fantoni R.**, Eur. Phys. J. B **86**, 286 (2013)
dx.doi.org/10.1140/epjb/e2013-40204-3
WOS:000321446200046
2-s2.0-84898868063

(40) **Fantoni R.**, Giacometti A., Maestre M. A. G., and Santos A., J. Chem. Phys. **139**, 174902 (2013)
dx.doi.org/10.1063/1.4827861
WOS:000326922300052
2-s2.0-84903365438

(41) **Fantoni R.** and Santos A., J. Chem. Phys. **140**, 244513 (2014)
dx.doi.org/10.1063/1.4884353
WOS:000338634200051
2-s2.0-84903710390

(42) **Fantoni R.** and Pastore G., J. Chem. Phys **141**, 074108 (2014)
dx.doi.org/10.1063/1.4892878
WOS:000340714600011
2-s2.0-84906544798

(43) **Fantoni R.**, Phys. Rev. E **90**, 020102(R) (2014)
dx.doi.org/10.1103/PhysRevE.90.020102
WOS:000341289600001
2-s2.0-84907237529

(44) **Fantoni R.** and Moroni S., J. Chem. Phys **141**, 114110 (2014)
dx.doi.org/10.1063/1.4895974
WOS:000342843200014
2-s2.0-84907247021

(45) **Fantoni R.** and Pastore G., Mol. Phys. **113**, 2593 (2015)
dx.doi.org/10.1080/00268976.2015.1061150
WOS:000362544000028
2-s2.0-84943582307

(46) **Fantoni R.** Giacometti A. and Santos A., J. Chem. Phys **142**, 224905 (2015)
dx.doi.org/10.1063/1.4922263
WOS:000356176600037
2-s2.0-84934994542

(47) **Fantoni R.**, Phys. Rev. E, **92**, 012133 (2015)
dx.doi.org/10.1103/PhysRevE.92.012133
WOS:000358439000001
2-s2.0-84938788007

(48) **Fantoni R.**, Eur. Phys. J. B **89**, 1 (2016)
dx.doi.org/10.1140/epjb/e2016-60917-9
WOS:000375216500004
2-s2.0-84957802219

(49) Alastuey A. and **Fantoni R.**, J. Stat. Phys. **163**, 887 (2016)
dx.doi.org/10.1007/s10955-016-1512-1
WOS:000374676000009
2-s2.0-84962025380

(50) **Fantoni R.**, Physica A **457**, 406 (2016)
dx.doi.org/10.1016/j.physa.2016.03.024
WOS:000376693600038
2-s2.0-84964504945

(51) **Fantoni R.**, J. Stat. Phys. **163**, 1247 (2016)
dx.doi.org/10.1007/s10955-016-1510-3
WOS:000375579300010
2-s2.0-84962010485

(52) **Fantoni R.**, J. Stat. Phys. **166**, 1334 (2017)
dx.doi.org/10.1007/s10955-016-1707-5
WOS:000395082700010
2-s2.0-85008514512

(53) **Fantoni R.**, Physica A **477C**, 187 (2017)
http://dx.doi.org/10.1016/j.physa.2017.02.064
WOS:000398873300017
2-s2.0-85014511370

(54) **Fantoni R.**, J. Stat. Phys. **168**, 652 (2017)
http://dx.doi.org/10.1007/s10955-017-1810-2
WOS:000405520600009
2-s2.0-85019677800

(55) **Fantoni R.** and Santos A., J. Stat. Phys. **169**, 1171 (2017)
https://doi.org/10.1007/s10955-017-1908-6
WOS:000415377700007
2-s2.0-85033500259

(56) **Fantoni R.**, J. Stat. Mech. P113101 (2017)
http://dx.doi.org/10.1088/1742-5468/aa9339
WOS:000414639900001
2-s2.0-85038631104

(57) **Fantoni R.**, J. Stat. Mech. P043101 (2018)
http://dx.doi.org/10.1088/1742-5468/aab690
WOS:000430327700001
2-s2.0-85046793056

(58) **Fantoni R.**, Int. J. Mod. Phys. C **29**, 1850028 (2018)
http://dx.doi.org/10.1142/S0129183118500286
WOS:000430039400009
2-s2.0-85044467252

(59) **Fantoni R.**, Int. J. Mod. Phys. C **29**, 1850064 (2018)
http://dx.doi.org/10.1142/S012918311850064X
WOS:000443597700002
2-s2.0-85049629270

(60) **Fantoni R.**, Physica A **515C**, 682 (2018)
https://doi.org/10.1016/j.physa.2018.09.107
WOS:000452941100061
2-s2.0-85054815484

(61) **Fantoni R.**, Physica A **524**, 177 (2019)
https://doi.org/10.1016/j.physa.2019.04.222
WOS:000476966100018
2-s2.0-85064919235

(62) **Fantoni R.**, Indian J. Phys. **95**, 1027 (2021)
https://doi.org/10.1007/s12648-020-01750-2
WOS:000543007600002
2-s2.0-85086857771

(63) **Fantoni R.**, J. Low Temp. Phys. **202**, 247 (2021)
https://doi.org/10.1007/s10909-020-02532-0
WOS:000574338800001
2-s2.0-85091785958

(64) **Fantoni R.**, Eur. Phys. J. B **94**, 63 (2021)
https://doi.org/10.1140/epjb/s10051-021-00078-y
WOS:000630734900001
2-s2.0-85103348101

(65) **Fantoni R.** and Klauder J. R., Phys. Rev. D **103**, 076013 (2021)
https://doi.org/10.1103/PhysRevD.103.076013
WOS:000648579600007
2-s2.0-85105542365

(66) **Fantoni R.**, J. Stat. Mech P083102 (2021)
https://doi.org/10.1088/1742-5468/ac0f69
WOS:000680165100001
2-s2.0-85111176604

(67) **Fantoni R.** and Klauder J. R., J. Stat. Phys. **184**, 28 (2021)
https://doi.org/10.1007/s10955-021-02818-x
WOS:000692401500001
2-s2.0-85111176723

(68) **Fantoni R.** and Klauder J. R., Phys. Rev. D **104**, 054514 (2021)
https://doi.org/10.1103/PhysRevD.104.054514
WOS:000704629900005
2-s2.0-85114892586

(69) **Fantoni R.**, Maestre M. A. G., and Santos A., J. Stat. Mech P103210 (2021)
https://doi.org/10.1088/1742-5468/ac2897
WOS:000711175900001
2-s2.0-85118939833

(70) **Fantoni R.**, Mol. Phys. **120**, 4 (2021)
https://doi.org/10.1080/00268976.2021.1996648
WOS:000712258400001
2-s2.0-85118339895

(71) **Fantoni R.** and Klauder J. R., Int. J. Mod. Phys. A **37**, 2250029 (2022)
https://doi.org/10.1142/S0217751X22500294
WOS:000769229500003
2-s2.0-85126738235

(72) **Fantoni R.** and Klauder J. R., Int. J. Mod. Phys. A **37**, 2250094 (2022)
https://doi.org/10.1142/S0217751X22500944
WOS:000811562100003
2-s2.0-85132227773

(73) **Fantoni R.**, Eur. Phys. J. C **82**, 843 (2022)
https://doi.org/10.1140/epjc/s10052-022-10807-x
WOS:000861486700003
2-s2.0-85135756093

(74) **Fantoni R.** and Klauder J. R., Phys. Rev. D **106**, 114508 (2022)
https://doi.org/10.1103/PhysRevD.106.114508
WOS:000906363700004
2-s2.0-85144770879

(75) **Fantoni R.**, J. Stat. Mech. 083103 (2023)
https://doi.org/10.1088/1742-5468/aceb54
WOS:001048913500001
2-s2.0-85169885952

(76) Klauder J. R. and **Fantoni R.**, Axioms **12**, 911 (2023)
https://doi.org/10.3390/axioms12100911
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(80) Klauder J. R. and **Fantoni R.**, Int. J. Mod. Phys. A **39**, 2450094 (2024)
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(81) **Fantoni R.**, The Physics Educator **6**, 2420004 (2024)
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(85) **Fantoni R.**, The Physics Educator **7** 2550002 (2025)
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WOS:
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(88) Klauder J. R. and **Fantoni R.**, Crystall Journal of Physics **1**, 1 (2025)
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(89) **Fantoni R.**, Gravitation and Cosmology **31**, 568 (2025)
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WOS:001620053000002
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(91) **Fantoni R.** and Klauder J. R., RRJ. Pure Appl. Phys. **13**, 1 (2025)
https://doi.org/10.4172/2320-2459.13.1.001
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(92) **Fantoni R.**, Mol. Phys. **124**, 4 (2026)
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WOS:001520107600001
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(93) **Fantoni R.**, Gravitation and Cosmology **32**, 314 (2026)
https://doi.org/10.1134/S020228932670012X
WOS:
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(94) **Fantoni R.**, Quantum Rep. **8**, 12 (2026)
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(95) **Fantoni R.**, Quantum Rep. **8**, 42 (2026)
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WOS:
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(96) **Fantoni R.**, Eur. Phys. J. B **99**, 89 (2026)
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WOS:
2-s2.0-105043596971

(97) **Fantoni R.**, Phys. Rev. B **114**, 185409 (2026)
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Books

(1) **Fantoni R.** "Regole di somma in un gas di elettroni stratificato", ISBN 978-889-101-539-6

(2) **Fantoni R.** "Classical liquids: exact results, integral equations theory, and Monte Carlo simulations", ISBN 978-889-101-543-3

(3) **Fantoni R.** "The Janus fluid, from a theoretical perspective", SpringerBriefs in Physics, (2013), ISBN 978-3-319-00406-8 (Print), 978-3-319-00407-5 (Online)
dx.doi.org/10.1007/978-3-319-00407-5

(4) **Fantoni R.** "Discoveries due to the observation of mathematics", Kindle Direct Publishing, (2022)

(5) Fantoni R. "Unifying Classical and Quantum Physics. How classical and quantum physics can pass smoothly back and forth", Springer Nature (2025), Print ISBN-10: 3031981227, Print ISBN-13: 978-3-031-98122-7, Electronic ISBN-13: 978-3-031-98123-4

Schools and Conferences

1994 Napoli CNR meeting. Poster on publication (1).

2002 National school of Matter Physics on "Fisica di base delle Nanostrutture e Calcolo ed Informazione Quantistica" (Torino Villa Gualino 9/9/2002-20/9/2002).

2002 School on "Fisica Statistica, Teoria della Probabilitá e Complessitá computazionale" (ICTP Trieste 26/8/2002-7/9/2002).

2003 XXII Fai della Paganella meeting: "Fisica Teorica e Struttura della Materia". Posters on publication (4) and on publication (5).

2004 XXIII Fai della Paganella meeting: "Fisica Teorica e Struttura della Materia". Poster on publication (6).

2004 "IX Convegno Nazionale di Fisica Statistica e dei Sistemi Complessi" (Parma 22-24 June). **Talk** on publication (8).

2004 IV Giovanni Paladin Memorial: "Statistical Mechanics, Chaos and Condensed Matter Theory" (Rome 22-24 September). Poster on publication (9).

2005 "X Convegno Nazionale di Fisica Statistica e dei Sistemi Complessi" (Parma 29-1 July). Poster on publications (9) and (10).

2005 "6th Liquid Matter Conference" (Utrecht, the Netherlands 2-6 July). Poster on publications (9) and (10).

2006 "31st Conference of the Middle European Cooperation in Statistical Physics" (Primošten, Croatia 23-26 April). Poster on publication (10).

2007 "Fluid phase behaviour and critical phenomena from liquid state theories and simulations" (CECAM, Lyon, France 12 July). **Talk** on publications (14) and (15).

2007 "Statphys23" (Genova, Italy 9-13 July). Poster on publications (14) and (15).

2008 "7th Liquid Matter Conference" (Lund, Sweden 27 June-1 July). Poster on publications (14) and (15).

2009 "Long-range Interactions in Classical and Quantum Physics" (Stellenbosch, South Africa 16-27 November). **Invited talk** on publication (17).

2010 "21st Chris Engelbrecht Summer School in Theoretical Physics" (Stellenbosch, South Africa 18-27 January).

2010 "35th Conference of the Middle European Cooperation in Statistical Physics" (Abbaye des Prómontrés, Pont-à-Mousson, France 15-19 March). Poster on publication (16), (18), and (20).

2010 "Statphys24" (Cairns, Australia 19-23 July). **Talk** on publication (17).

2010 **Invited talk** at the University of Extremadura in Badajoz on publications (16), (18), (20), (21), and (24).

2011 "22st Chris Engelbrecht Summer School in Theoretical Physics" (Stellenbosch, South Africa 19-30 January).

2011 "Equilibration and Equilibrium 2nd Stellenbosch Workshop on Statistical Physics" (Stellenbosch, South Africa 7-18 March). **Invited talk** on publication (24).

2011 "Workshop on Frontiers in Ultracold Fermi Gases" (Trieste, ICTP, Italy 6-10 June). Poster on publications (33) and (39).

2011 "8th Liquid Matter Conference" (Vienna, Austria, 6-10 September). Poster on publications (23) and (24).

2011 "National Institute for Theoretical Physics of South Africa (NITheP)" (Stellenbosch, South Africa, 21 Sptember). **Invited talk** on publication (23).

2011 "International Workshop on Ultracold Molecules" (Stellenbosch, South Africa, 7-11 November).

2012 "XCVIII Congresso Nazionale SIF" (Napoli, Italy, 17-21 September). **Talk** on publication (30).

2013 "38st Conference of the Middle European Cooperation in Statistical Physics" (Trieste, Italy, 25-27 March). Poster on publications (34) and (39).

2013 "Italian National Conference on Condensed Matter Physics" (Milano, Italy, 9-13 September). **Talk** on publication (41).

2013 "XCIX Congresso Nazionale SIF" (Trieste, Italy, 23-27 September). Abstract on Publication (36) and (40).

2013 **Invited talk** at the University of Extremadura in Badajoz on publications (41).

2014 "Sigma-Phi-2014" (Rhodes, Greece, 7-11 July). **Chairman** and **talk** on publication (43) and (44).

2015 "Italian National Conference on Condensed Matter Physics (FisMat 2015)" (Palermo, Italy, 28 September-2 October). **Talk** on publication (46).

2017 **Invited talk** at the University of Extremadura in Badajoz on publications ....

2017 "Workshop on Understanding Quantum Phenomena with Path Integrals: From Chemical Systems to Quantum fluids and Solids — (smr 3131)" (Trieste, ICTP, 3-7 July). Partecipant.

2017 "10th Liquid Matter Conference" (Ljubljana, Slovenia, 17-21 July). Poster on publications (55)

2017 "Italian National Conference on Condensed Matter Physics (FisMat 2017)" (Trieste, Italy, 1-5 October). **Talk** on publication (55)

2018 "6th World Congress and Expo on Nanotechnology and Material Science" (Valencia, Spain, 16-18 April). **Invited Talk** on publication (59).

2025 "Memorial Conference in Honor of John R. Klauder" (Gainesville, USA, 15 February). **Invited Talk** on publications from the trilogy (84), (85), (86).

Supervision of students

- Cosupervision of the Ph.D. thesis of Joris W. O. Salari in the department of polymer chemistry of the University of Technology of Eindhoven in the Netherlands. Supervisor: Prof. Bert Klumperman.
- Cosupervision of the Ph.D. thesis of Miguel A. G. Maestre in the physics department of the University of Extremadura in Spain. Supervisor: Prof. Andrés Santos.

Scientific software available to the research international community

- Development of a Mathematica notebook which evaluates the radial distribution functions for binary mixtures of nonadditive hard spheres, according to the method described in R. Fantoni and A. Santos, "Nonadditive hard-sphere fluid mixtures. A simple analytical theory", Phys. Rev. E **84**, 041201 (2011).
- Development of a Gibbs Ensemble Monte Carlo algorithm to determine the phase properties of a ionic fluid of non-additive hard spheres, according to the method described in R. Fantoni and G. Pastore "Monte Carlo simulation of the nonadditive restricted primitive model of ionic fluids: Phase diagram and clustering", Phys. Rev. E **87**, 052303 (2013).
- Development of a new Quantum Monte Carlo algorithm to determine the phase properties of a quantum fluid of bosons, according to the method described in R. Fantoni and S. Moroni, "Quantum Gibbs ensemble Monte Carlo", J. Chem. Phys. **141**, 114110 (2014).

Other activities

- Editorial Board Member of Deep Science Publishing from 20 April 2026
- Editorial Board Member of Mathematics (MDPI) from 24 November 2023 to 24 November 2025
- Editorial Board Member of American Journal of Modern Physics (AJMP) from 18 April 2023 to 1 September 2028

- Editorial Board Member of Journal of Information Analysis from 25 April 2023 to 25 April 2026
- Editorial Board Member of Academic Journal of Physics Research from 11 July 2024
- reviewer for the American Mathematical Society (AMS), the American Physical Society (APS), the American Institute of Physics (AIP), the Institute of Physics (IOP), the Royal Society of Chemistry (RSC), the European Physical Society (EPS), the J. High Energy Phys., Gravitat. Cosmol.
- member of SAIP, SIF, EPS
- ranked C3 by the National Research Foundation of South Africa (Division of Research Development) in 2011
- Responsibility in projects of supercomputing: Use of the facilities of NCSA at Urbana-Champaign, Illinois, U.S.A. (1995-2000); Use of the facilities of CHPC at Cape Town, South Africa (2010-2012); Use of the facilities of CINECA at Bologna, Italy (2011-present).